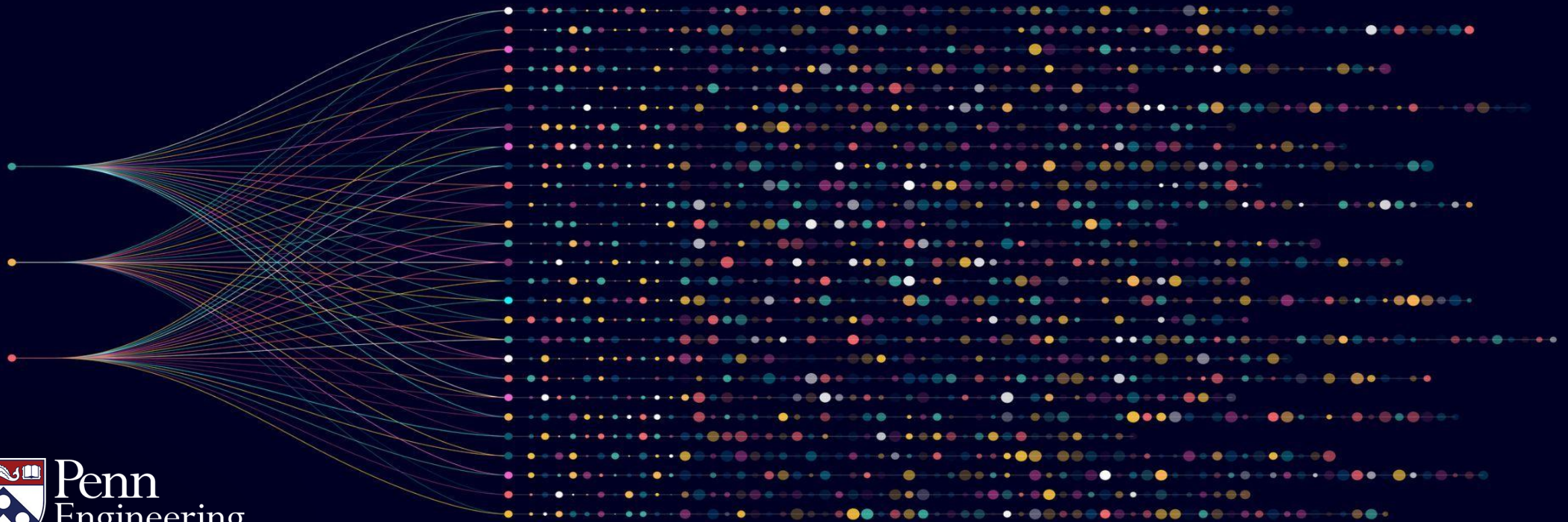


Algorithms for Big Data

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Pitfall of Deterministic Algorithms



Objectives: Computing Large Averages

- Randomized algorithms for large averages
- Why deterministic algorithms cannot produce accurate approximations

Defining the Computational Problem.

Input: $X_1, \dots, X_n \in \mathbb{R}$ and $\epsilon \geq 0$

Goal: Output $\hat{S}$ which satisfies $\left| \hat{S} - \frac{1}{n} \sum_{i=1}^n X_i \right| \leq \epsilon.$

Bad news: Deterministic algorithm must make linearly many queries

What is randomization?

- Randomized algorithm output random variables
 - Algorithm “flips coins” internally, specific output is inherently unpredictable
- For approximate guarantees with sublinearly many queries, randomness is unavoidable

Why is randomness necessary?

Input variables:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1
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A deterministic algorithm (which does not use any randomness) is a rule for selecting queries

Goal: two inputs with different averages which look the same to a deterministic algorithm

We construct two examples to “fool” algorithm

Input variables:

0	0			0					0						0
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Assume a deterministic algorithm making 5 queries.

- When query position i , make that 0

Average

Possible Inputs:

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0

0	0	1	1	0	1	1	1	1	0	1	1	1	1	1	0
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Precise bounds for the “fooling set” argument

- Fix a value of n and a deterministic algorithm making q queries
- Two inputs:
 - All 0's input — Average = 0
 - The q queries are 0, rest are 1's — Average = $1 - q/n$
- Best case scenario: $\frac{1}{2}(1 - q/n) \leq \epsilon$.
- Conclusion: queries must be at least $n(1 - 2\epsilon)$

The pitfall of deterministic algorithms

- Deterministic algorithms commit to a fixed action
- If the number of queries is small, there are many values which were not observed
- There are two different inputs which are indistinguishable to the algorithm, but have significantly different answers